



Hydrogen fluoride emissions from lithium-ion batteries during induced thermal runaway via in situ laser spectroscopy

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ABSTRACT

Improved understanding of hydrogen fluoride (HF) emissions from lithium-ion battery fires, including the temporal dynamics, is needed to optimize fire response and protection. Due to the high polarity of HF and its associated surface adsorption and reactivity, most traditional sensing methods are prone to error and slow response due to issues with sampling or surface interactions. To address these limitations, an in situ tunable diode laser absorption spectrometer is developed to achieve real-time measurements of HF emissions during dynamic battery fires with a temporal resolution of milliseconds, and with detection limits of single parts per million along with several orders of magnitude of dynamic range. The laser spectrometer is used in situ to perform measurements near the fire source so that the fire dynamics and the transient behavior of HF emissions can be more accurately characterized. Thermal runaway and fire/explosion conditions of model 18650 lithium-ion batteries are simulated in a conical radiative heater, and HF measurements are performed online via an optical access port in the effluent exhaust. By varying the radiative heating flux of the conical heater and the initial state of charge of the batteries, different characteristics of the safety venting and thermal runaway behavior of lithium-ion batteries and the corresponding emissions of toxic HF gas are measured. These findings provide valuable insights into the dynamics of lithium-ion battery fires and will aid in the development of strategies to mitigate their associated risks.

1. Introduction

The increasing reliance on lithium-ion (Li-ion) batteries in consumer electronics, electric vehicles, and other energy storage systems has raised significant concerns regarding their safety. The risks associated with Li-ion batteries stem from their high energy density and the potential for thermal runaway, a self-propagating process initiated by exothermic reactions within the cell. This process can lead to hazardous gas emissions, fire, and explosions. The progression of thermal runaway can generally be divided into several distinct stages. The initial heat-up stage occurs when external mechanical, thermal, and electrical abuse causes the battery temperature to rise. As external heating progresses, the battery transitions to the onset of a self-heating stage (T_o), triggering exothermic reactions between the electrolytic fluid and charged anode, leading to further temperature increases until the battery reaches the rapid thermal runaway threshold (T_r). In the rapid thermal runaway stage, the complete breakdown of electrolytic barriers causes highly exothermic reactions with oxidizers inside the battery or ambient air, driving an abrupt and exponential temperature rise until reaching a local maximum temperature (T_m) [1,2]. An additional intermediate stage, known as the safety venting stage, has

also been reported in certain studies on cylindrical and prismatic Li-ion cells [3,4]. In this stage a burst of gas emission occurs before the onset of thermal runaway due to a rupture of the battery's venting disk at a lower temperature (T_v), a safety design feature intended to relieve internal pressure. Battery thermal runaway is influenced by multiple factors, including state of charge (SOC), battery chemistry, and environmental conditions. The release of gas emissions during the various stages of the battery progression to thermal runaway can serve as an important indicator for safety and fire management.

Previous studies have extensively examined temperature evolution and gas emissions during thermal runaway using thermocouple measurements and gas sampling-based techniques, with notable limitations. First, gas sampling methods for ex situ analysis, such as Fourier Transform Infrared Spectroscopy (FTIR) and gas chromatography, suffer from slow response times and low temporal resolution relative to the fast transient events of a Li-ion battery fire [3,4]. Second, the remote location of the analytical instruments and requisite sampling leads to adsorption and surface reaction losses of less stable species along the sampling lines (on top of the additional measurement latency and sample dispersion). This introduces measurement inaccuracies, particularly

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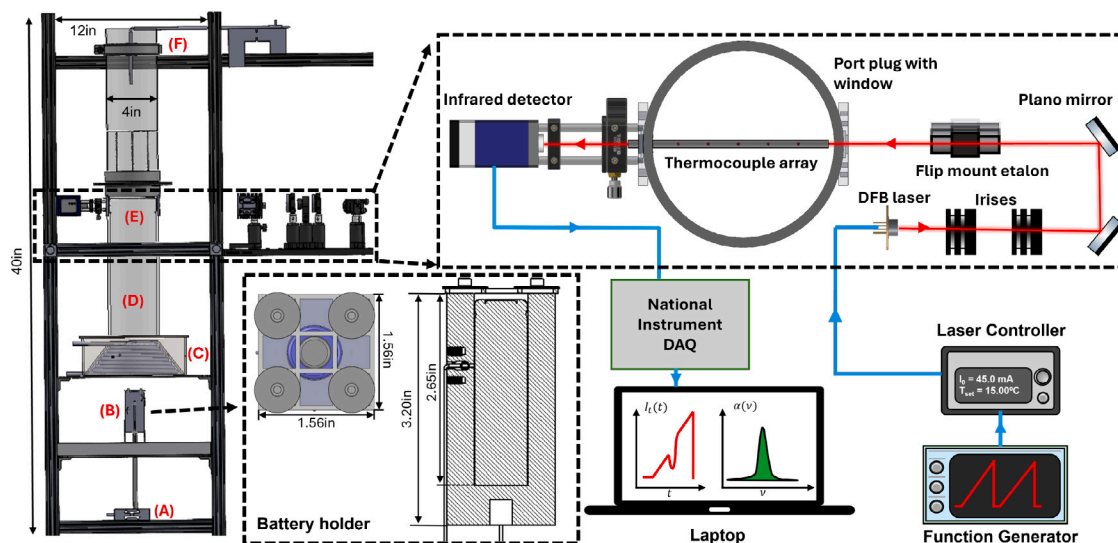
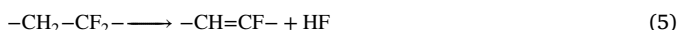


Fig. 1. Schematic of the experimental setup. Left: Radiative heater facility includes (A) single-point load cell, (B) battery holder, (C) conical heating element, (D) gas exhaust duct, (E) gas test section with LAS setup, (F) pitot-thermocouple velocimeter. Bottom Middle: Top view and side view cross-section of battery holder with a testing Li-ion battery inserted. Top Right: LAS optical components and electronics.

in the case of reactive gas species such as hydrogen fluoride (HF).

Hydrogen fluoride is among the most toxic byproducts of Li-ion battery thermal runaway, with health impacts including pulmonary inflammation, dermal burns, and systemic toxicity [5]. When Li-ion batteries experience external abuses such as external heating, hydrogen fluoride is primarily generated from the decomposition of the electrolyte salt (LiPF_6) and the fluorinated polymer binder (PVDF), which degrade at different temperature ranges [6]. These decomposition pathways result in distinct HF emission characteristics due to different electrochemical reactions and associated activation energies [7–9], with the key global reaction equations in (1)–(5). As indicated, the high electronegativity of fluorine effectively abstracts hydrogen from water and other species to form HF.



Due to the high polarity of hydrogen fluoride and its tendency to adhere to and react with surfaces, accurately quantifying HF emissions remains a significant challenge in fire safety research. Most previous studies have relied on indirect or delayed measurement methods, leading to potentially errant measurements of HF concentrations [10,11]. To overcome these limitations, this study employs in situ laser absorption spectroscopy (LAS) to enable online, high time-resolution measurements of HF emissions during Li-ion battery fires. LAS offers highly quantitative in situ optical detection with kHz time resolution, eliminating sampling line losses and measurement latency. We use a radiative heater to induce controlled thermal runaway conditions for cylindrical 18650 Li-ion batteries under varying SOC conditions. The high-speed in situ HF measurements obtained in this study provide critical insights into HF emission dynamics, improving the understanding of Li-ion battery fire hazards and informing fire protection strategies.

2. Methods

2.1. Experimental facility and instrumentation

To investigate Li-ion battery fires under controlled thermal abuse conditions, a radiative heating experiment was constructed (Fig. 1.

Left). A single cylindrical Li-ion battery (Sanyo NCR 18650 GA) was positioned vertically in a custom-designed battery holder (Fig. 1(B)) and subjected to blackbody radiation via a conical heating element (Fig. 1(C)). The battery state of charge preparation followed the methods and procedures described in [2] to ensure consistency and repeatability across tests. The battery holder was mounted on a single-point load cell via a ceramic connecting rod (Fig. 1(A)) to continuously monitor battery mass loss during testing. To track the thermal response of the battery, two spring-loaded thermocouples were installed in direct contact with the battery surface through the battery holder walls. During the experiment, gaseous battery emissions from venting and thermal runaway were directed through an exhaust duct (Fig. 1(D)), which included a designated test section that is optically accessible (Fig. 1(E)), such that gas properties could be measured in situ using a laser absorption spectrometer (Fig. 1 Top Right) and the temperature of the gas was determined using a thermocouple array (TC array) with five thermocouples installed 0.5 inches above the laser path. As the sample gas moves further along the exhaust duct, it passes through a flow conditioner, ensuring a more uniform flow profile before reaching the pitot-thermocouple velocimeter (Fig. 1(F)) that provides an exhaust gas flow rate measurement, allowing for the total mass of HF emitted to be determined. All measurement signals, except for the LAS detector output, were recorded using two synchronized data acquisition devices (DATAQ DI-2008) at a sampling rate of 33 Hz. The LAS detector signal was acquired separately using a high-speed data acquisition device (NI-PXI 6115) with a sampling rate of 400 kHz, enabling high-resolution spectral analysis.

2.2. Controlled battery heating rate

A coiled 3.55-meter-long Incoloy 800 conical heater (5 kW maximum power), controlled via surface temperature feedback, was employed to deliver radiative heating flux to the test battery. The surface heating flux on the top plate of the battery holder was measured by a heat flux gauge. Given the vertical orientation of the battery, and relatively small exposed area, direct radiative heating was somewhat secondary, and the primary heat transfer to the test battery occurred through thermal conduction via the larger battery holder, which was heated by the radiation source and internal reflections from radiation shielding in the test chamber. To further quantify the effective thermal energy gain to the test battery, the rate of battery temperature increase during the linear heat-up phase, prior to the onset of safety

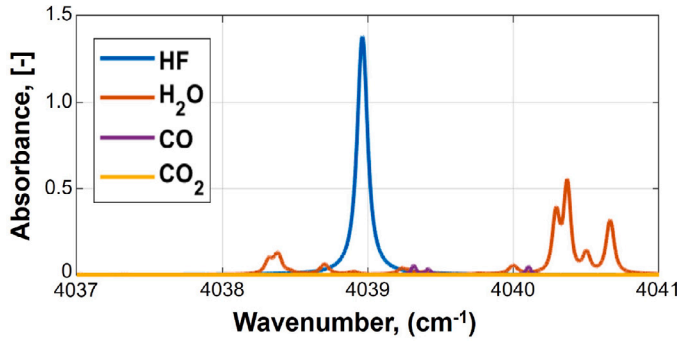


Fig. 2. Simulated target transitions of major gas products during battery fire at gas temperature of 1800 K, including HF ($X_{\text{HF}} = 1\%$), H_2O ($X_{\text{H}_2\text{O}} = 15\%$), CO ($X_{\text{CO}} = 30\%$), and CO_2 ($X_{\text{CO}_2} = 40\%$).

venting, was also measured. This temperature gradient served as an indirect metric for assessing the net energy absorbed by the battery under different heating conditions. Three different heater temperature setpoints were selected to produce distinct battery surface temperature ramp rates. These conditions were designed to align with the three thermal runaway propagation behaviors identified in [12], and enabled a systematic investigation of how heating rate influences HF emissions dynamics and thermal runaway behaviors.

2.3. HF laser absorption spectroscopy

The mole fraction of HF in the test section was measured using a tunable diode laser absorption spectroscopy (TDLAS) setup. A distributed feedback (DFB) diode laser (Norcada NL2475) was used to target the R(1) rovibrational transition of HF at 4038.965 cm^{-1} in the fundamental band of HF [13] over a 4 inch optical path length across the exhaust duct above the heater. A laser controller (Arroyo 6305) was used to set the laser chip temperature and the injection current, which was modulated with a 1 kHz linear ramp function from a function generator (Rigol DG1032Z) to achieve wavelength tuning across the absorption line of HF. After passing through the sample gas, the transmitted laser beam is detected and conditioned by a VIGO photovoltaic detector (MIP-DC-50M-F-M8) and recorded on an external data acquisition device (NI-PXI 6115) at a 400 kHz sampling rate.

In this study, the LAS sensor was scanned in wavenumber from 4037.5 cm^{-1} to 4041 cm^{-1} , which fully captures the selected HF spectral line and neighboring features as shown in Fig. 2. Within this spectral range, several water vapor absorption transitions from $E'' = 1581\text{--}4728 \text{ cm}^{-1}$ are also active at high temperatures, providing an alternative means for optical gas temperature measurement based on the Boltzmann intensity distribution. Other major combustion gas species, such as CO and CO_2 , are shown to have negligible interference with the target HF line, per Fig. 2. During Li-ion battery fires, a certain amount of volatile hydrocarbons are generated [14], but their C-H vibrational modes do not fall in the spectral range of this study. One representative laser intensity scan highlighting the resolved lines during a battery fire test is shown in Fig. 3 (top). The corresponding absorbance spectrum was calculated via $\alpha_v = -\ln(I_t/I_0)_v$ where I_t and I_0 represent the transmitted and incident intensities of the laser light, respectively. The absorbance spectrum was then integrated using $\int_v \alpha_v = S(T)X_iPL$ to determine the HF mole fraction, where $S(T)$ is the spectral linestrength, P is the gas pressure, L is the laser path length, and X_i is the gas mole fraction. To recover the integrated absorbance areas, the spectrum for each scan was fitted using Voigt lineshape profiles and line parameters from the HITRAN database as shown in Fig. 3 (bottom), allowing for the simultaneous retrieval of gas temperature, HF mole fraction, and water vapor mole fraction [15,16].

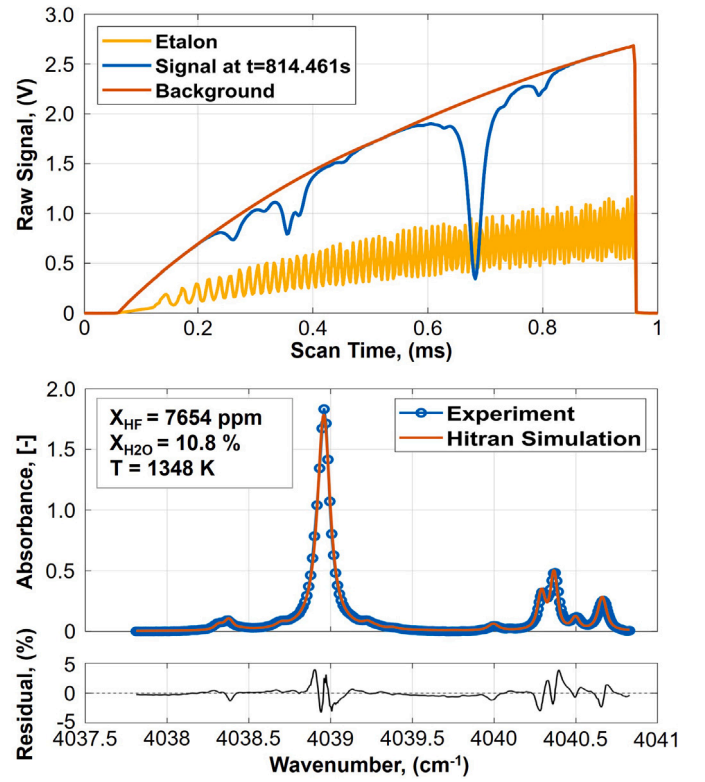


Fig. 3. Top: Instantaneous light absorbance signal during battery fire event. Bottom: Spectral fit of the instantaneous signal with the HITRAN database for gas properties measurement.

2.4. Exhaust gas measurement

The exhaust gas temperature and velocity were measured to characterize gas dynamics and mass flow rate during the battery fire events. The flow velocity was determined using a pitot-thermocouple velocimeter, while the gas temperature was recorded by a thermocouple next to the pitot tube. The calculation of exhaust gas velocity was based on $v_{\text{gas}} = K \sqrt{\frac{2\Delta P_{\text{gas}}}{\rho_{\text{gas}}(T_{\text{gas}})}}$, where ΔP_{gas} is the pressure difference between stagnation and static pressure, $\rho_{\text{gas}}(T_{\text{gas}})$ is the temperature-dependent gas density, and the K factor is an empirically derived correction coefficient, calibrated to be 1.025. The mass flow rate was then determined using $\dot{m}_{\text{gas}} = \rho_{\text{gas}}(T_{\text{gas}}) \cdot v_{\text{gas}} \cdot A_{\text{tube}}$.

The exhaust gas temperature (T_{gas}) measurement was corrected based on the energy balance of the thermocouple bead according to [17]. In this study, potential heat losses due to conduction and radiation are expected to be minimal, owing to the small size of the thermocouple bead and its supporting tube, as well as the relatively small temperature difference between the bead and the thermally insulated surrounding walls. Therefore, the updated correction relation from [17] only counts for the convection effect using $T_{\text{gas}} - T_{\text{TC}} = \frac{\rho_{\text{TC}} C_{p,\text{TC}} D}{4h} \frac{\partial T_{\text{TC}}}{\partial t}$ where ρ_{TC} and $C_{p,\text{TC}}$ are the density and specific heat capacity of bead material, respectively, and h is the convective heat transfer coefficient. The value of h was evaluated using Whitake's empirical correlation [18].

3. Results

In this study, nine battery thermal abuse tests with three different battery states of charge (100%, 50%, and 0%) were conducted at three levels of radiative heating power yielding distinct levels of equilibrium battery temperature gain rates of approximately 21.3 K/min, 12.8

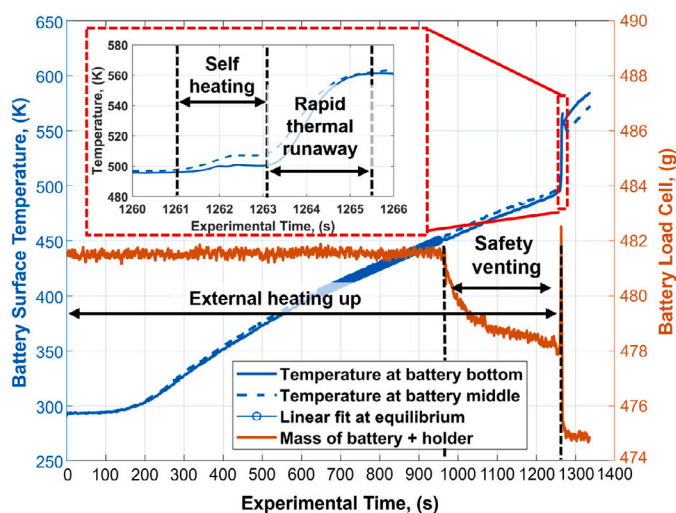


Fig. 4. Temperature and mass measurement of the battery (and holder) over the experiment period with the illustration of four stages during a Li-ion battery fire.

K/min, and 7.6 K/min corresponding to the measurement in different thermal runaway propagation processes in [12]. To facilitate discussion of the typical battery fire events and gas emission dynamics, a representative experiment with a partially charged battery (50% SOC) under a heating flux of 36 kW/m² is highlighted in Figs. 4–6 and discussed in the following subsections.

3.1. Battery temperature and mass evolution

At the onset of heating, the battery temperature initially rises slowly due to the thermal inertia of the holder (and battery) and uneven heat distribution. After this initial phase (around 300 s in Fig. 4), the temperature rise gradient becomes semi-constant during a linear heat-up stage, until reaching $T_o = 498$ K at 1261 s, marking the onset of the self-heating stage, in which the battery temperature increased faster than the externally driven rate. Approximately two seconds after onset, the rapid thermal runaway threshold was reached ($T_r = 504$ K). This final stage was characterized by an accelerated mass loss due to electrolyte vaporization and venting as well as a sharp temperature increase that reached a local maximum of $T_m = 561$ K, accompanied by the violent rupture of the battery's internals and the expulsion of hot gases leading a sharp spike downward force on the battery load cell. Additionally, a significant precursor mass loss event was observed at $t = 965.7$ s and $T_o = 455$ K, corresponding to the safety venting stage reported in previous studies [3,4]. During this stage, the venting disk ruptured, releasing gaseous decomposition products prior to thermal runaway. Notably, the accumulated mass loss during venting was comparable to, or even higher than, that which was observed during thermal runaway stages, but this mass loss occurred at a slower rate.

3.2. HF emission dynamics and battery fire evolution

To further analyze the fire dynamics, real-time HF mole fraction measurements were obtained using in situ laser absorption spectroscopy (LAS). A representative temperature and HF mole fraction evolution in the exhaust over the experiment period is shown in Fig. 5 (Middle), alongside time-synchronized frames from a video recording of the experiment to better visualize the fire progression (Top). Two distinct spikes in HF emissions were detected with mole fraction value over 1%, corresponding to the onset of safety venting (t_1) and the onset of thermal runaway (t_5). During the first fire event, a visible flame was present, accompanied by significant HF emissions. Although this flame weakened over time (t_2), it stabilized for an extended period (minutes),

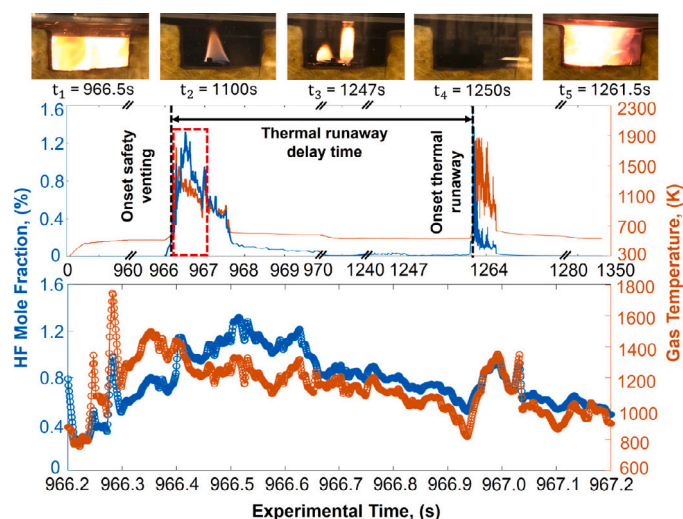


Fig. 5. HF emission and fire evolution. Top: Instantaneous picture frames of test battery during fire events. Middle: HF mole fraction and temperature of emission gas at test section over the experiment period. Bottom: zoomed window of gas mole fraction and temperature during dynamic period of safety venting.

suggesting that hot electrolyte vapors were continually released and oxidized. As the fire progressed, the weak flame became unstable (t_3) and temporarily extinguished (t_4). The battery fire was reignited about 15 s later (t_5), leading to a second, more intense fire event. This secondary fire was significantly stronger with visible hot metal sparks, and coincided with a higher HF emission peak and a sharper temperature increase. Unlike the second fire event, which required higher temperatures to breach additional internal barriers, the fire in the safety venting stage was not detected by the battery surface temperature measurement, similar to past studies [2]. This is presumably because the flame in the venting stage was well-ventilated and did not cause the self-heating of the battery itself, making it difficult to capture using thermocouple measurements on the battery surface. While battery surface temperature measurements were insufficient to fully capture the fire dynamics, gas-phase temperature measurements provide a more direct method for characterizing the fire intensity in the test section. However, accurately capturing these fast-changing gas temperatures presents its own set of challenges. A common challenge in fast dynamic temperature measurements using thermocouples is their slow response time as mentioned in the previous section, often leading an underestimate of gas temperature even with the thermocouple correction approach [17]. To overcome this limitation, LAS thermometry was employed for more accurate measurement of gas temperature (when the water lines become sufficiently strong above 600 K) and HF gas mole fraction as shown in Fig. 5 (Bottom), benefiting from a response time over two orders of magnitude faster than the thermocouples (1 ms in LAS versus seconds in TC array).

3.3. Hydrogen fluoride production

While assessments based on mole fraction can provide insight into local gas concentrations, they are heavily influenced by ventilation conditions and the size of the surrounding environment, making direct comparisons across different fire scenarios challenging. In contrast, mass-based quantification provides a scalable and environmentally-independent measurement of toxic emissions, offering a more reliable approach to hazard assessment. Accumulated HF emissions on a mass basis were calculated from HF mole fraction and exhaust gas mass flowrate measurement. As shown in Fig. 6, The total HF production during the safety venting stage was about 124 mg which is about three times larger than the HF production during the thermal runaway stage

Table 1
18650 Li-ion battery thermal abuse experiment test results.

dT/dt (K/min)	SOC (%)	Onset temp.		Battery WL		TR delay (s)	Peak HF emi.		Accumulated HF emi.			
		SV (K)	TR (K)	SV (g)	TR (g)		SV (%)	TR (%)	SV (mg)	TR (mg)	Total ¹ (mg/Wh)	Total ² (%)
23.6	100	447	512	2.2	25.3	233	2.74	6.69	196	54	22.4	0.91
21.7	50	458	540	2.3	18.4	256	3.13	1.63	174	23	17.7	0.95
18.7	0	491	595	4.6	0.8	356	8.51	0.10	182	18	17.9	3.70
14.0	100	432	481	2.1	26.5	248	1.80	3.73	122	18	12.6	0.49
12.6	50	455	498	3.3	3.6	295	1.52	1.54	124	40	14.2	2.41
11.7	0	476	681	5.5	0.4	NA	2.58	0.05	117	12	11.6	2.19
7.68	100	427	478	3.4	25.2	320	0.08	4.66	69	27	8.65	0.34
7.50	50	437	493	3.5	7.7	456	0.30	1.82	31	21	4.59	0.46
7.56	0	453	658	5.3	0.8	NA	0.35	0.02	51	3.7	4.87	0.89

Total¹ is the total HF emission weight from safety venting (SV) and thermal runaway (TR) over full battery energy.

Total² is the total HF emission over the total battery weight loss (WL).

NA in thermal runaway (TR) delay time indicates the onset of TR triggered at higher heating flux.

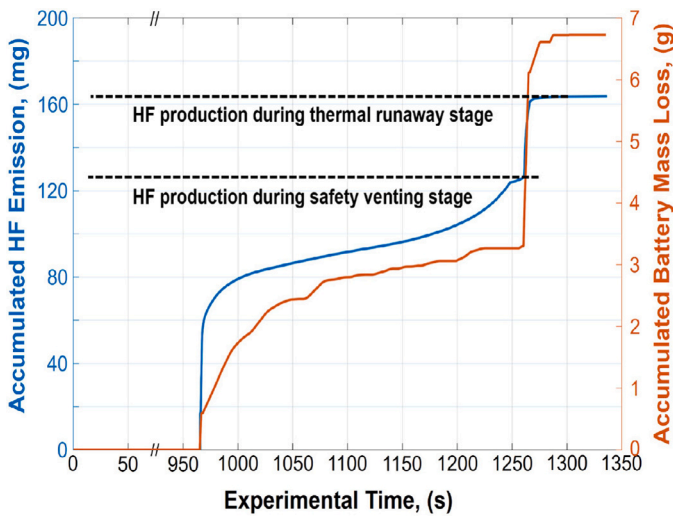


Fig. 6. Accumulated total HF mass production and the battery mass loss over the experiment period.

(40 mg in this specific case). Notably, a significant amount of HF was generated during the latter period of the gas venting stage (from t_2 to t_5 in Fig. 5). In contrast with the larger amount of HF produced in the safety venting stage, the battery mass loss experienced a more pronounced spike during the thermal runaway stage.

4. Discussion

Battery fire characteristics from the test series, including key emission metrics, are summarized in Table 1. The comparative results include initiation temperature of safety venting and thermal runaway, thermal runaway delay time with respect to the onset time of safety venting, peak HF emission, stage-specific HF mass production, and HF production normalized by battery energy capacity and battery mass loss. Trends in the dataset with respect to battery heating rate and state of charge are discussed in this section.

4.1. Onset temperature

The onset temperatures of battery fire stages of safety venting and thermal runaway ranged from 415 K to 492 K and from 461 K to 681 K, respectively. Both ranges fall within the findings reported in prior work [2,3]. With an increase of SOC level, both onset temperature values decrease due to the greater availability of internal electrochemical

energy. With a higher heating flux, both onset temperatures measured on the battery surface increase likely due to greater thermal gradients between the battery surface and core, providing that each electrochemical reaction ignition temperature is consistent across different external heating fluxes. The gap between the safety venting and thermal runaway onset temperatures becomes smaller at higher SOC levels as more energy is available to quickly transition from safety venting to thermal runaway.

4.2. Thermal runaway delay time

Based on the fast response and high time-resolution of the in situ HF LAS detection system, distinct spikes in emissions are observed to correspond to safety venting and thermal runaway, and thus a delay time between the two events can be well-quantified. This highlights the potential for a real-time HF spectrometer to provide forewarning of the more explosive thermal runaway event. At a constant heat flux, the thermal runaway delay time with respect to safety venting decreases with increasing SOC. This is because higher SOC levels contain more electrochemical energy, which accelerates the reactions leading to thermal runaway. At a constant SOC, increasing the heating flux also reduces the delay time, as it causes the battery temperature to rise more quickly and reach the reaction threshold sooner. A special case at 0% SOC was observed that at applied heating flux values of 27 and 36 kW/m², the 0% SOC battery only exhibited safety venting fire without progressing to thermal runaway. This suggests that there was insufficient energy input — both electrochemical (internal) and thermal flux (external heat) — to trigger the full runaway reactions. Thermal runaway was eventually observed at 0% SOC when a higher heating flux (53 kW/m²) was applied.

4.3. Battery weight loss

The battery weight loss characteristics generally align with different major electrochemical reactions at varying activation temperatures, and appear consistent with prior work [6]. These physical trends in mass loss are discussed briefly here to provide context for subsequently discussed HF emissions trends. In general, as battery temperature increases, electrolyte decomposition and partial combustion occur first. The extent of this first-stage combustion is influenced by the onset temperature of safety venting. In the thermal runaway stage, higher temperatures can lead to non-reaction solid particle expulsion associated with over-pressurization of the battery structure, causing significant additional weight loss. Accordingly, in this study, across all SOC levels and heating fluxes, weight loss during thermal runaway is higher than during safety venting. Between 50% and 100% SOC, venting-related weight loss is similar, but thermal runaway involves more intense

solid-particle expulsions at 100% SOC, resulting in greater weight loss. At 0% SOC, venting-related weight loss is larger than in the other SOC cases. With low electrochemical energy, combustion relies more heavily on external heating, resulting in more complete hydrocarbon fuel combustion with delayed over-pressurization. Due to the absence of significant energy to help drive reactions and explosive behavior, the thermal runaway stage causes minimal additional weight loss at 0% SOC.

4.4. HF emission concentrations

The peak concentration of HF emissions reached several percent in mole fraction, which is two orders of magnitude higher than values reported in existing literature [10]. This discrepancy is likely due to the high temporal resolution of LAS measurements, which accurately captures transient emission peaks, as well as the direct in situ measurement approach, which eliminates potential gas losses caused by adsorption or surface reaction inside the sampling lines. These factors contribute to a more accurate quantification of HF emissions compared to traditional gas sampling approaches and gas analysis techniques. HF measurement uncertainty was estimated to be 10% using the methods of [19]. A higher HF mole fraction peak was observed at higher SOC and higher heating flux presumably due to more internal electrochemical energy and stronger external thermal energy gain accelerating the HF production reactions. Comparing the safety venting and thermal runaway stages, the instantaneous HF mole fractions were either similar or higher during the thermal runaway stage, likely due to less total gas being generated. As will be discussed in the following sub-section, the metric of mole fraction alone in the evaluation of effluent toxicity can be misleading, especially in large scale fire accidents or with well ventilated conditions.

4.5. HF mass production

To quantify fire toxicity and exposure risk within a given spatial domain or volume, mass-based production from fire effluents is more relevant than instantaneous concentration [5], given that mass production of fire effluents is very dynamic. In this study, the total HF mass production from the batteries ranged from approximately 50 mg to 250 mg (Table 1). Given the full charge energy rating of 11.55 Wh, the total HF generation per unit energy capacity in this study ranged from 4.6 to 22 mg/Wh. This variation aligns with or falls near previously reported values [3]. The total HF mass generation per unit mass loss of the battery was also calculated in a wide range of 0.34% to 3.7% depending on the amount of total battery material loss from the thermal runaway stage. Notably, as shown in the example test in Fig. 6, and across other conditions as seen in Fig. 7, HF emissions production is consistently higher during the safety venting stage than during thermal runaway. This is because the primary source of HF is electrolyte decomposition, which occurs at relatively low temperatures. In contrast, HF generated during thermal runaway results mainly from the pyrolysis of fluorinated protective materials, which requires much higher temperatures. As also shown in Fig. 7, the total HF emission mass production (at a constant SOC) increases with heating rate while being weakly dependent on state-of-charge. The total HF emission does not vary significantly across different SOC because the electrolyte and fluoride binder material are conserved in weight. However, lower external heating flux levels would lead to a decrease in total HF generation likely because the intermediate fluorinated gas species such as PF_5 , POF_3 , generated from electrolyte decomposition reactions (1) and (3) were emitted before their subsequent hydrolysis reactions (2) and (4) which require a higher temperature.

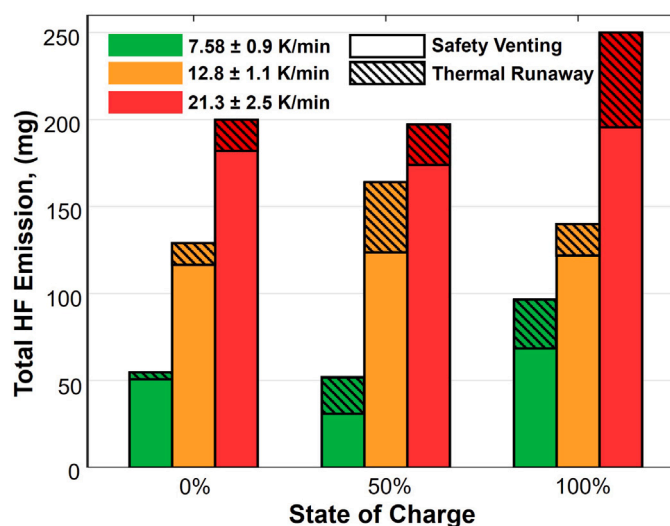


Fig. 7. HF mass production in safety venting stage and thermal runaway stage with different heating gain rate, and state of charge level.

5. Conclusions

This study examined Li-ion battery thermal abuse under controlled heating conditions, with a particular focus on HF emissions. Under varying heat input power and battery state of charge, we quantified the dynamics of thermal runaway and gas venting through time-resolved measurement of battery temperature, battery mass loss, gas temperature, and HF emissions. The latter element represents a novel implementation of in situ laser absorption spectroscopy for high-speed time-resolved HF emission measurements, which have proven difficult to acquire with traditional instrumentation. The high temporal resolution (1 kHz) of the LAS system in this study enables online monitoring of rapid off-gassing events, capturing transient emission spikes, and quantifying emissions production at different stages of the battery fire. This novel approach provides more accurate emission quantification and fire dynamics characterization.

The time-resolved HF emission measurements helped elucidate the two distinct emissions events during a Li-ion battery fire: the safety venting stage and the rapid thermal runaway stage. While the thermal runaway stage has been extensively studied due to its explosion hazards, this work illustrates the significance of the venting stage, as it can contribute to a substantial portion of toxic HF emissions. The study reveals that HF release during the early safety venting stage can significantly exceed emissions observed during thermal runaway. Overall, the findings offer valuable insights into battery fire dynamics and toxic gas exposure risks that are important for developing safer energy storage technologies and fire response protocols.

Novelty and significance statement

This study presents the novel application of in situ tunable diode laser absorption spectroscopy (LAS) for highly time-resolved quantification of hydrogen fluoride (HF) emissions during induced lithium-ion battery thermal runaway, addressing gaps in fire safety knowledge due largely to insufficient instrumentation to quantify hydrogen fluoride. By synchronizing LAS with controlled radiative heating experiments across varying states of charge (SOC), this study characterizes two distinct periods of HF emissions associated with safety venting and thermal runaway stages during a lithium-ion battery fire, and quantifies the cumulative HF emissions from each period. This integrated experimental facility and diagnostic method provides unique information for numerical model development and validation as well as new insights for fire response and management.

CRediT authorship contribution statement

Yi Yan: Designed and performed this research experiment, analyzed the data, and wrote the paper. **Nicolas S.B. Jaeger:** Designed the experiment, provided suggestions on data analysis, and reviewed and revised the manuscript. **R. Mitchell Spearrin:** Designed the experiment, provided comments and suggestions on experiment performance and data analysis, reviewed and revised the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] O. Willstrand, M. Pushp, P. Andersson, D. Brandell, Impact of different Li-ion cell test conditions on thermal runaway characteristics and gas release measurements, *J. Energy Storage* 68 (2023) 107785.
- [2] A.W. Golubkov, D. Fuchs, J. Wagner, H. Wilsche, C. Stangl, G. Fauler, G. Voitic, A. Thaler, V. Hacker, Thermal-runaway experiments on consumer Li-ion batteries with metal-oxide and olivin-type cathodes, *RSC Adv.* 4 (7) (2014) 3633–3642.
- [3] L. Lin, O.A. Ezekoye, Time-resolved characterization of toxic and flammable gases during venting of Li-ion cylindrical cells with current interrupt devices, *J. Loss Prev. Process. Ind.* 94 (2025) 105488.
- [4] K.C. Abbott, J.E. Buston, J. Gill, S.L. Goddard, D. Howard, G. Howard, E. Read, R.C. Williams, Comprehensive gas analysis of a 21700 $\text{Li}(\text{Ni}_{0.8}\text{Co}_{0.1}\text{Mn}_{0.2})\text{O}_2$ cell using mass spectrometry, *J. Power Sources* 539 (2022) 231585.
- [5] Committee on Toxicology and Board on Environmental Studies and Toxicology, Acute Exposure Guideline Levels for Selected Airborne Chemicals, Volume 4, The National Academies Press, Washington, DC, 2004.
- [6] H. Li, Q. Duan, C. Zhao, Z. Huang, Q. Wang, Experimental investigation on the thermal runaway and its propagation in the large format battery module with $\text{Li}(\text{Ni}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3})\text{O}_2$ as cathode, *J. Hazard Mater.* 375 (2019) 241–254.
- [7] H. Yang, G. Zhuang, P. Ross Jr., Thermal stability of LiPF_6 salt and Li-ion battery electrolytes containing LiPF_6 , *J. Power Sources* 161 (2006) 573–579.
- [8] T. Kawamura, S. Okada, J.-i. Yamaji, Decomposition reaction of LiPF_6 -based electrolytes for lithium ion cells, *J. Power Sources* 156 (2006) 547–554.
- [9] F. Larsson, P. Andersson, P. Blomqvist, A. Loren, B.-E. Mellander, Characteristics of lithium-ion batteries during fire tests, *J. Power Sources* 271 (2014).
- [10] F. Larsson, P. Andersson, P. Blomqvist, B.-E. Mellander, Toxic fluoride gas emissions from lithium-ion battery fires, *Sci. Rep.* 7 (1) (2017) 10018.
- [11] P. Ribière, S. Grugeon, M. Morcrette, S. Boyanov, S. Laruelle, G. Marlair, Investigation on the fire-induced hazards of Li-ion battery cells by fire calorimetry, *Energy Env. Sci.* 5 (1) (2012).
- [12] J. Jeevarajan, C. Lopez, S. Azam, S. Kinyon, Thermal runaway propagation testing study for safe transportation of lithium-ion cells and batteries, in: AIAA Propulsion and Energy 2019 Forum, Am. Inst. Aeronaut. Astronaut., Indianapolis, IN, 2019.
- [13] C.I. Sanders, K.A. Oberlander, R.M. Spearrin, Polytetrafluoroethylene (PTFE) burn characteristics and toxicant formation in an oxidizer cross-flow via laser absorption tomography, *PROCI* 40 (2024) 105394.
- [14] S. Koch, A. Fill, K.P. Birke, Comprehensive gas analysis on large scale automotive lithium-ion cells in thermal runaway, *J. Power Sources* 398 (2018) 106–112.
- [15] R.K. Hanson, R.M. Spearrin, C.S. Goldenstein, Spectroscopy and Optical Diagnostics for Gases, Springer International Publishing, Cham, Switzerland, 2016.
- [16] I.E. Gordon, L.S. Rothman, R.J. Hargreaves, R. Hashemi, et al., The HITRAN2020 molecular spectroscopic database, *J. Quant. Spectrosc. Radiat. Transf.* 277 (2022) 107949.
- [17] K. Kar, S. Roberts, R. Stone, M. Oldfield, Instantaneous exhaust temperature measurements using thermocouple compensation techniques, in: SAE 2004 World Congress & Exhibition, Society of Automotive Engineers, Detroit, MI, 2004.
- [18] S. Whitaker, Forced convection heat transfer correlations for flow in pipes, past flat plates, single cylinders, single spheres, and for flow in packed beds and tube bundles, *AIChE J.* 18 (1972) 361–371.
- [19] A.P. Nair, D.D. Lee, D.I. Pineda, J. Kriesel, W.A. Hargus, J.W. Bennewitz, S.A. Danczyk, R.M. Spearrin, MHz laser absorption spectroscopy via diplexed RF modulation for pressure, temperature, and species in rotating detonation rocket flows, *Appl. Phys. B* 126 (138) (2020).